EXPERIMENT 8: CONFIGURING USER MANAGEMENT AND ROLE-BASED ACCESS CONTROL (RBAC) IN A ZOHO CREATOR APPLICATION

Aim

To configure user management, roles, profiles, field-level security, and role-based access control (RBAC) in Zoho Creator for the Automated Corporate Talent Acquisition and Interview Tracking System to ensure data privacy, governance, and structured operational security across organizational stakeholders.

Theory

Role-Based Access Control (RBAC) and User Management in enterprise cloud applications enforce authorization boundaries so that users access only the data and functionality required for their operational roles. In Zoho Creator, security management combines organizational hierarchies, profile permission sets, field-level security, dynamic Deluge rules, and portal access controls.

Key user management and RBAC concepts implemented in this module include:

- **User Categorization & Portal Users:** Zoho Creator distinguishes between internal workspace users (Administrators, Developers, Recruiters) and external Portal Users (Job Applicants / Candidates), allowing controlled self-service candidate access without exposing administrative workflows.
- **Role Hierarchy & Data Visibility:** Roles reflect the organizational reporting structure (e.g., HR Director -> Hiring Manager -> Senior Recruiter -> Technical Interviewer). Data sharing rules inherit visibility hierarchically, granting managers automatically scoped access to records created by subordinates.
- **Profiles & Permission Sets:** Profiles define granular module-level CRUD (Create, Read, Edit, Delete) permissions across forms, reports, and dashboards. Custom profiles (e.g., HR Admin Profile, Recruiter Profile, Interviewer Profile) prevent unauthorized actions across recruitment modules.
- **Field-Level Security (FLS):** Field-level access control restricts visibility and edit permissions for sensitive fields (e.g., Expected Salary, Offer Amount, Confidential Interview Ratings, Internal HR Notes) based on the active user's assigned profile.
- **Dynamic Security via Deluge Scripting:** Deluge scripts dynamically enforce contextual access rules at runtime using system variables like zoho.loginuser, zoho.loginuserid, and zoho.adminuserid to filter report data or conditionally disable form elements.

Procedure

1. Log in to Zoho Creator and open the existing Automated Corporate Talent Acquisition and Interview Tracking System application.

2. Navigate to Application Settings and select Users & Permissions under the Security & Access section.

3. Configure organizational Roles establishing a top-down hierarchy: HR Director (Top Level), HR Recruiter (Mid Level), and Technical Interviewer (Operational Level).

4. Create custom Profiles to enforce granular form and module permissions:
• HR Admin Profile: Full access (Create, Read, Edit, Delete) across all forms, reports, and administrative tools.
• Recruiter Profile: Create/Edit permissions for Candidate Applications, Job Requisitions, and Interview Schedules, with Delete disabled.
• Technical Interviewer Profile: Read-only permission for assigned candidate details and Edit permission restricted to Interview Evaluation forms.

5. Set up Field-Level Security (FLS) on the Candidate Application form to mark fields like Expected Salary and Internal HR Remarks as hidden or read-only for the Technical Interviewer profile.

6. Configure Data Sharing Rules and Criteria-Based Access Control so that Technical Interviewers can only view candidate records assigned to them in the Interview Schedule.

7. Enable Customer Portal access for external candidate registration, permitting applicants to submit profiles and view real-time status updates while isolating internal HR forms.

8. Implement On Load Deluge scripts on evaluation forms using zoho.loginuser to dynamically hide confidential scoring fields or restrict entry to authorized evaluation panel members.

9. Perform RBAC testing by logging in under different user personas (HR Director, Recruiter, Technical Interviewer, External Candidate) to verify restricted form navigation, report filters, and field visibility.

10. Inspect security execution and audit logs in Zoho Creator to verify that unauthorized access attempts are correctly blocked.

Sample Deluge Security Scripts & Dynamic Access Control

Below are reference Deluge code implementations configured and tested for dynamic role-based control and field security:

1. On Load Workflow: Dynamic Field Masking Based on User Role / Email

```
// On Load Event - Candidate Application Form
// Hide sensitive financial and internal fields from non-HR users
currentUser = zoho.loginuser;
if (currentUser != zoho.adminuserid)
{
    hide Expected_Salary;
    hide Offered_CTC;
    hide Internal_HR_Notes;
    info "Sensitive HR fields masked for user: " + currentUser;
}
```

2. Record-Level Evaluation Access Control

```
// On Load Event - Interview Evaluation Form
// Restrict interviewers from modifying evaluations assigned to other panellists
if (input.Interviewer_Email != zoho.loginuser)
```

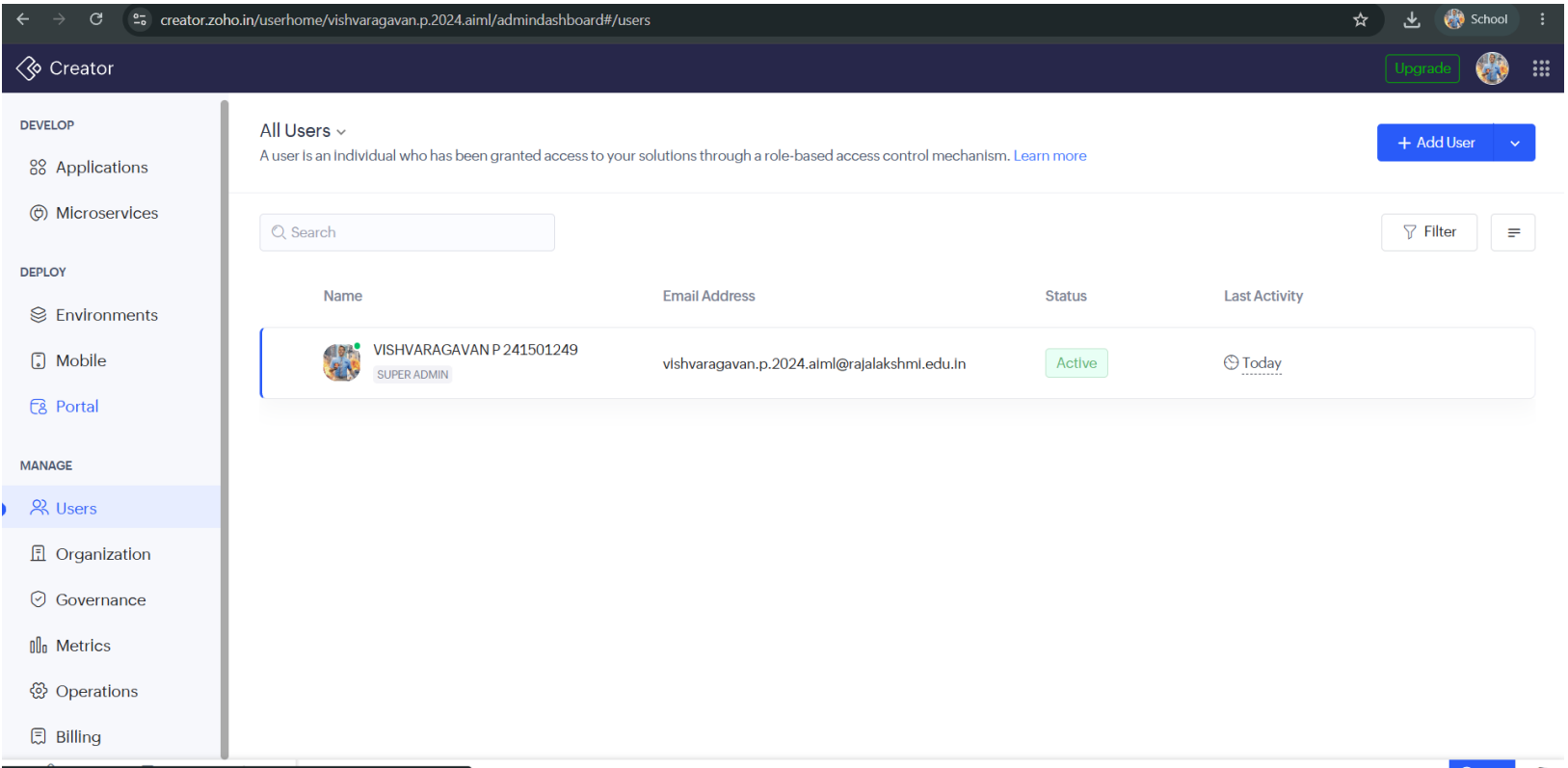
```
{
    disable Rating_Score;
    disable Recommendation_Status;
    alert "You are only authorized to complete evaluations assigned to your
account.";
}
```

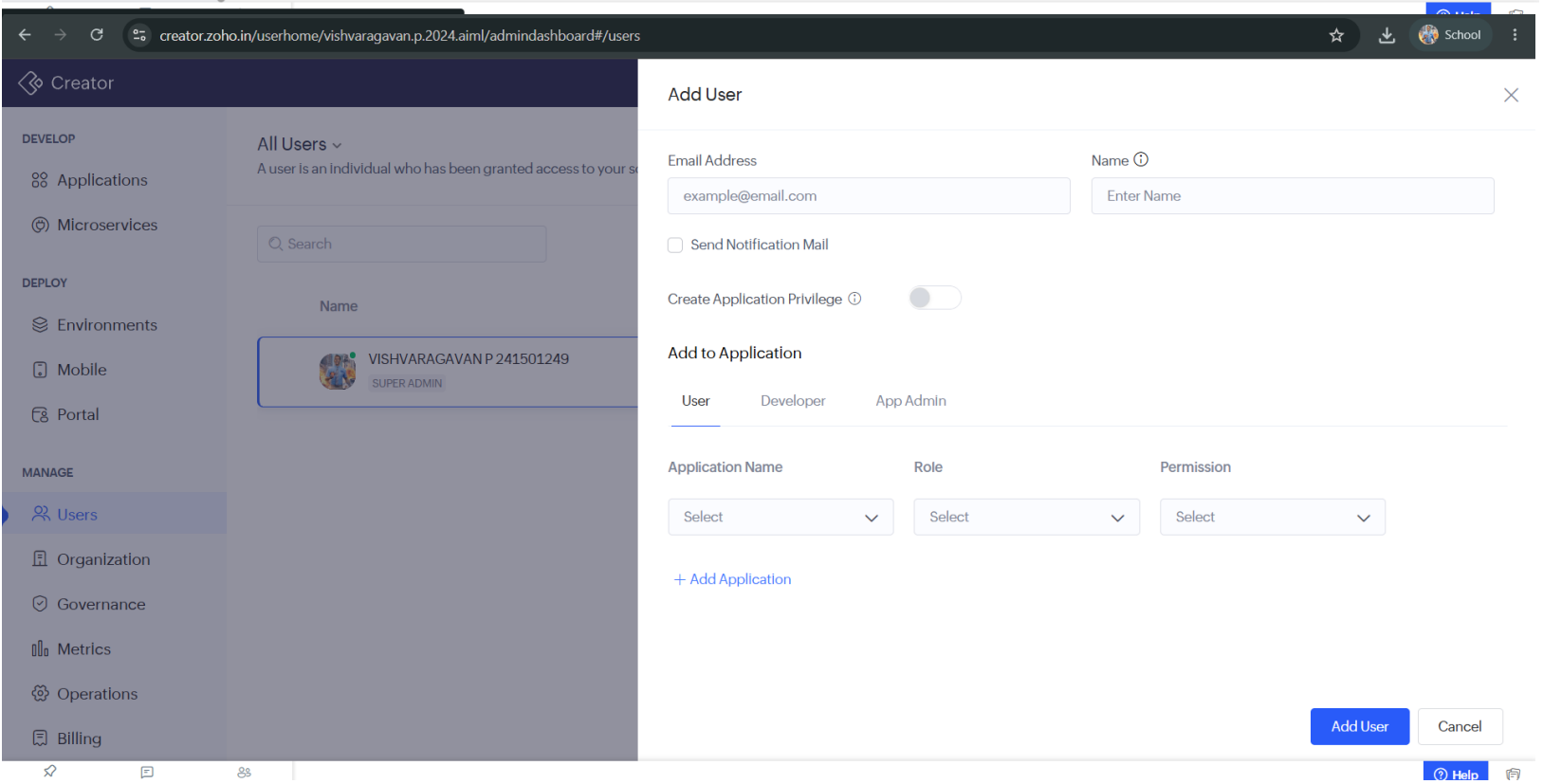
3. On User Input Workflow: Enforcing Candidate Status Lockdown

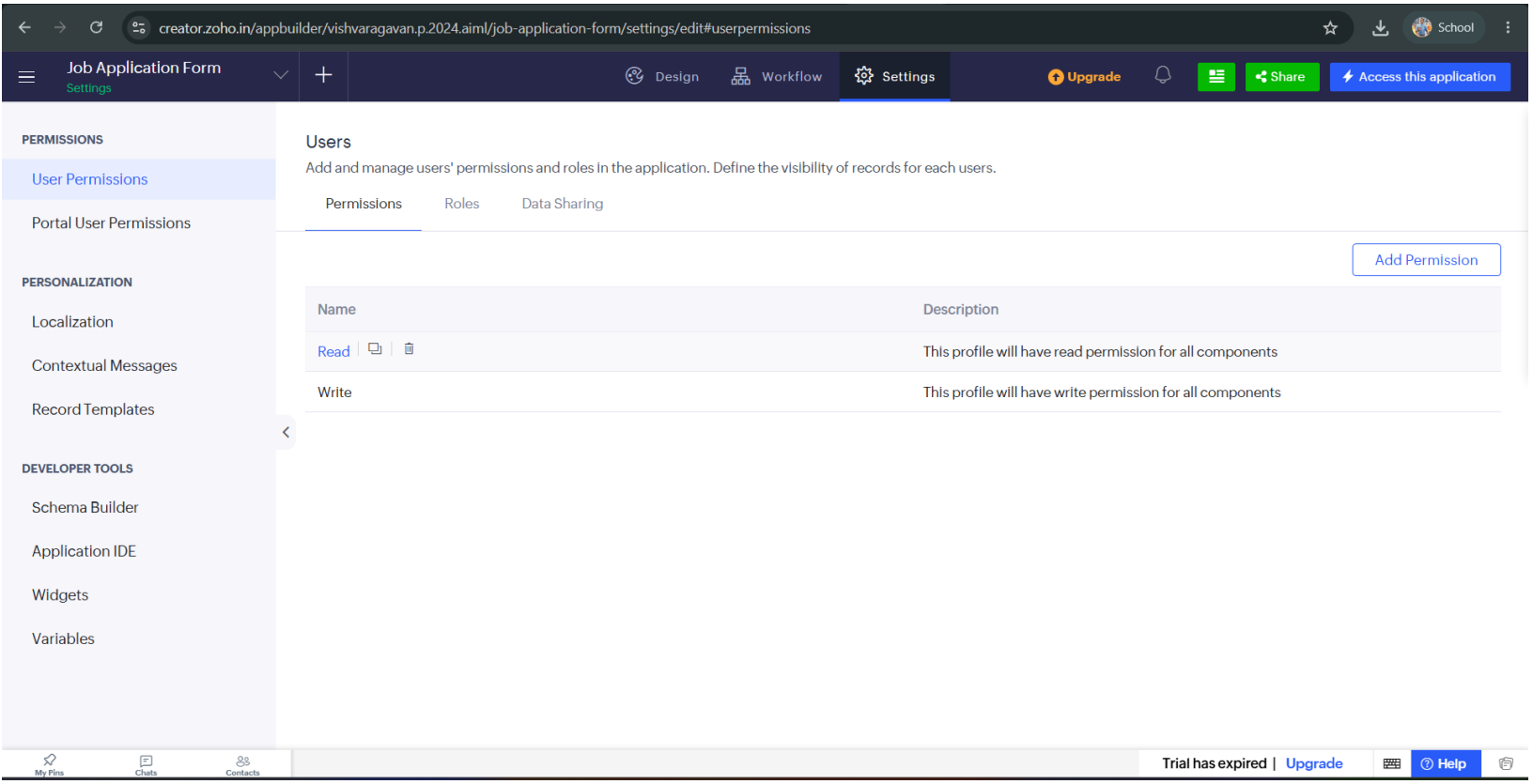
```
// On User Input Event - Candidate Status Field
// Restrict Status changes to authorized recruiters only
if (input.Application_Status == "Selected" || input.Application_Status ==
"Rejected")
{
    if (zoho.loginuser != zoho.adminuserid)
    {
        alert "Only HR Administrators have permission to finalize selection or
rejection decisions.";
        input.Application_Status = oldvalue.Application_Status;
    }
}
```

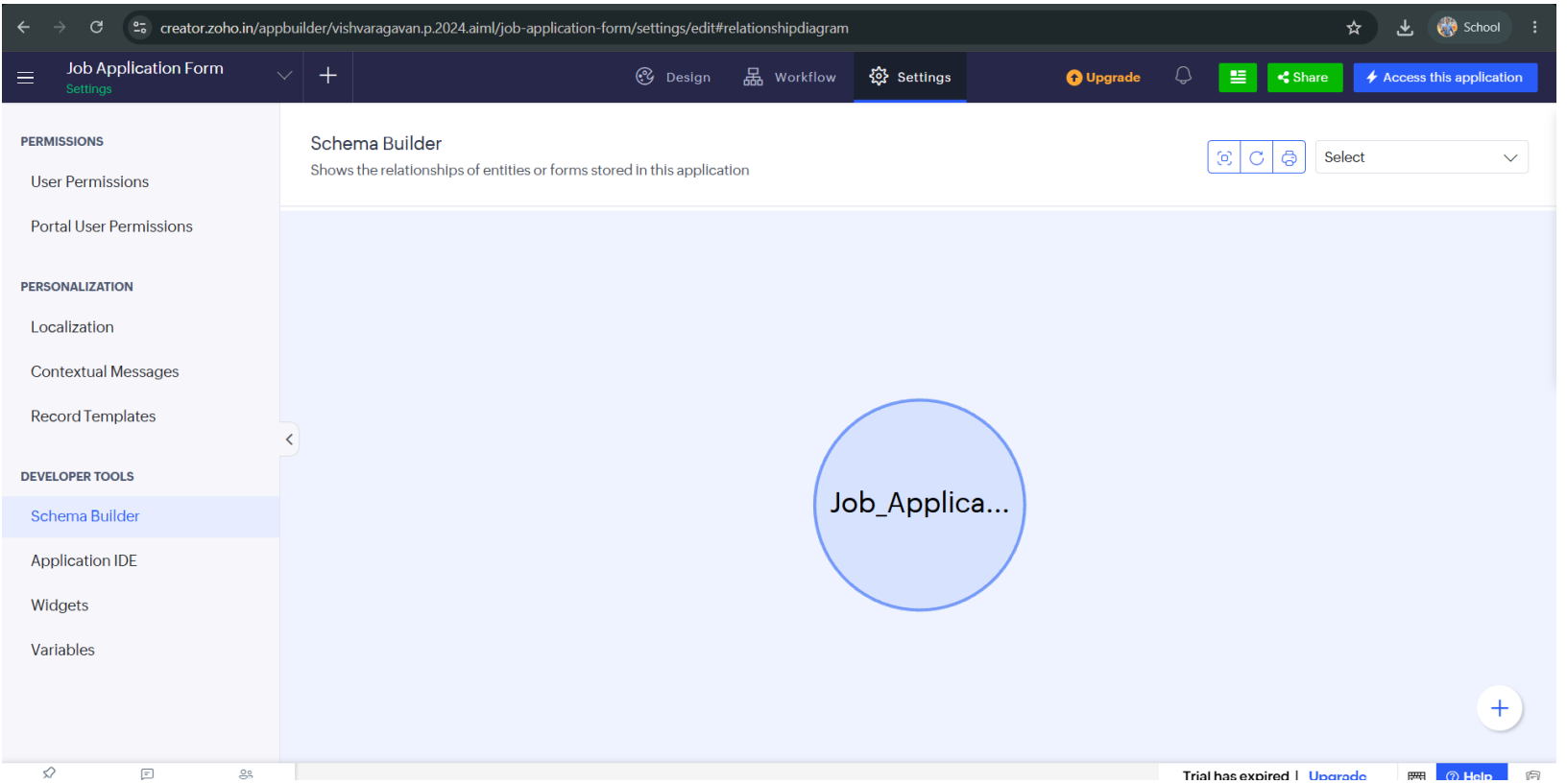
4. Dynamic Report Scoping for Technical Interviewers

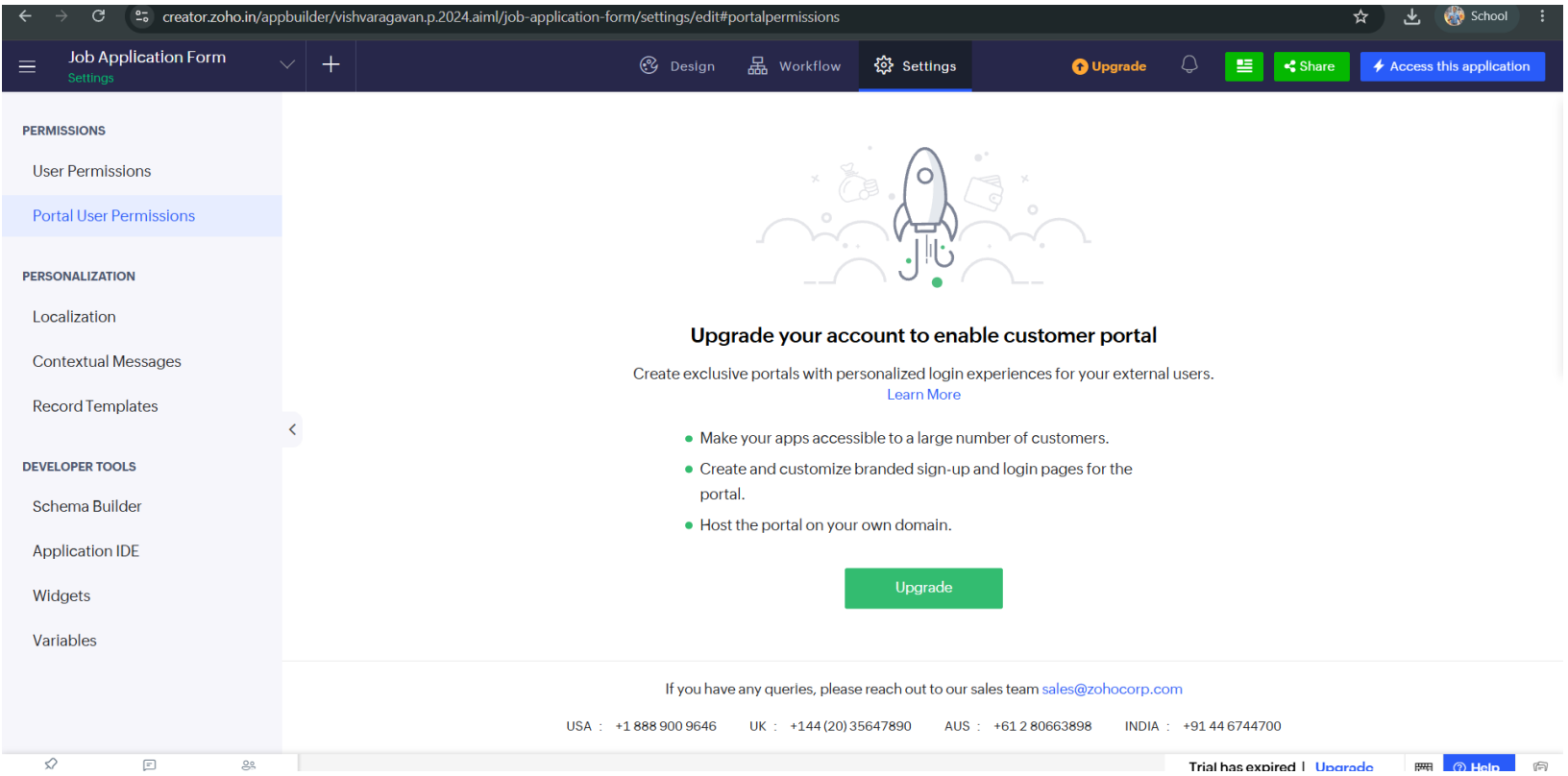
```
// Deluge Criteria Script for Custom Assigned Interviews Report View
// Filters report view dynamically based on logged-in user
Assigned_Interviews = Interview_Schedule[Assigned_Interviewer_Email ==
zoho.loginuser];
info "Total interviews scheduled for logged-in interviewer: " +
Assigned_Interviews.count();
```


creator.zoho.in/userhome/vishvaragavan.p.2024.aiml/admindashboard#/users
School
Creator
Upgrade
DEVELOP
Applications
Microservices
DEPLOY
Environments
Mobile
Portal
MANAGE
Users
Organization
Governance
Metrics
Operations
Billing
All Users
A user is an individual who has been granted access to your solutions through a role-based access control mechanism. Learn more
+ Add User
Search
Filter
Name
Email Address
Status
Last Activity
VISHVARAGAVAN P 241501249
SUPER ADMIN
vishvaragavan.p.2024.aiml@rajalakshmi.edu.in
Active
Today


creator.zoho.in/userhome/vishvaragavan.p.2024.aiml/admindashboard#/users
School
Creator
DEVELOP
Applications
Microservices
DEPLOY
Environments
Mobile
Portal
MANAGE
Users
Organization
Governance
Metrics
Operations
Billing
All Users
Search
Name
VISHVARAGAVAN P 241501249
SUPER ADMIN
Add User
Email Address
example@email.com
Name
Enter Name
Send Notification Mail
Create Application Privilege
Add to Application
User
Developer
App Admin
Application Name
Role
Permission
Select
Select
Select
+ Add Application
Add User
Cancel
Help


creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/settings/edit#userpermissions
School
Job Application Form
Settings
Design
Workflow
Settings
Upgrade
Share
Access this application
PERMISSIONS
User Permissions
Portal User Permissions
PERSONALIZATION
Localization
Contextual Messages
Record Templates
DEVELOPER TOOLS
Schema Builder
Application IDE
Widgets
Variables
Users
Add and manage users' permissions and roles in the application. Define the visibility of records for each users.
Permissions
Roles
Data Sharing
Add Permission
Name
Description
Read
This profile will have read permission for all components
Write
This profile will have write permission for all components
My Pins
Chats
Contacts
Trial has expired | Upgrade
Help


creator.zoho.in/appbuilder/vishvaragavan.p.2024.aiml/job-application-form/settings/edit#relationshipdiagram
School
Job Application Form
Settings
Design
Workflow
Settings
Upgrade
Share
Access this application
PERMISSIONS
User Permissions
Portal User Permissions
PERSONALIZATION
Localization
Contextual Messages
Record Templates
DEVELOPER TOOLS
Schema Builder
Application IDE
Widgets
Variables
Schema Builder
Shows the relationships of entities or forms stored in this application
Select
Job_Applica...
Trial has expired | Upgrade
Help



Result

Thus, user management and Role-Based Access Control (RBAC) were successfully configured in Zoho Creator for the Automated Corporate Talent Acquisition and Interview Tracking System. Roles, profiles, field-level security, data-sharing rules, and portal access controls were thoroughly tested and verified to ensure secure, role-specific data privacy across HR administrators, recruiters, technical interviewers, and job applicants.